

Computer Applications Lab

0907331

LabSheet8: Matplotlib

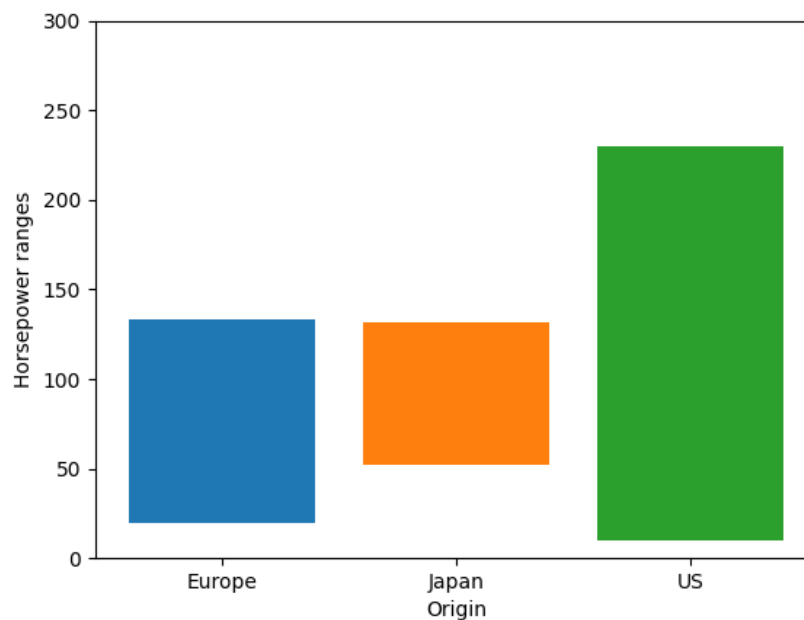
Answer the following questions on the **sales**, **cars**, **Uploads** and **movies** datasets. The datasets are available on MS teams.

Notes

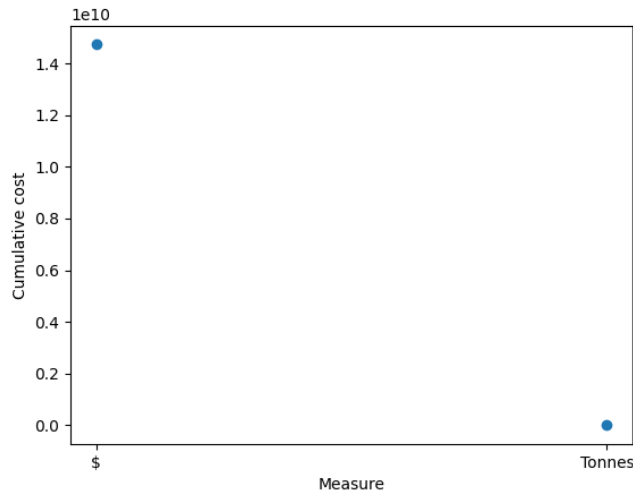
1. Grading will be based on the correctness, clarity and the completeness of the figure elements like labels, title and legend if needed.
2. The figures should appear in the questions order
3. Fill the missing values in each column with the average value for that column.

Tasks

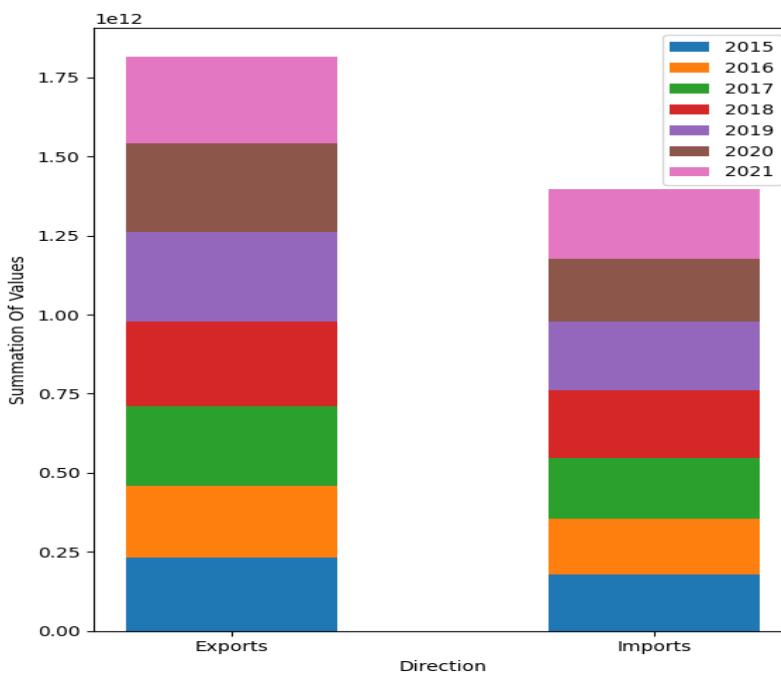
1. Using the Sales.xlsx dataset answer the following
 - a. Draw a stackplot that shows the sales per month
 - b. Draw a pie chart that shows the percentage of sales for each product in August
 - c. Find the Sugar sales for each quarter and draw it as a bar plot
2. Using the cars.xlsx dataset answer the following
 - a. Draw a pie plot that shows the percentage of the cars displacement in the ranges (0-100, 100-200, 200-300, 300-400, over 400)
 - b. Draw a stacked bar plot that shows the ranges of horsepower for each country as shown.



3. Using the Uploads.csv (uploads effects of COVID 19 on trade) dataset answer the following
 - a. Draw a scatter plot that shows the sum of cumulative cost for the Milk powder, butter, and cheese exports that done in 27/11/2015 with value more than 9000 as shown



- b. Draw a stack bar plot that shows the summation of values for the two trade directions Exports and Imports for the year from 2015 to 2021 as shown below.



4. Using the movies.xls dataset answer the following
 - a. Draw a pie chart for the movies produced by Germany. The pie chart shows the distribution of the movies based on the language.
 - b. Draw a bar chart that shows the number of movies produced by Spain, France, Canada and China in each year between 2011 and 2016 inclusive.

- c. Draw a scatter plot that shows the relation between the budget and the IMDB score for movies produced by France. The size of the dot should be based on the duration of the movie and the color based on the first letter of the movie.
- d. Draw a pie chart that shows the percentage of the movies for each Genre. Note: if the movie is Action and War then you count it in the Action group and the War group